System for Artificial Intelligence

Introduction | Fall 2026

Siyuan Feng
Shanghai Innovation Institute

OUTLINE

01 Why AI Systems

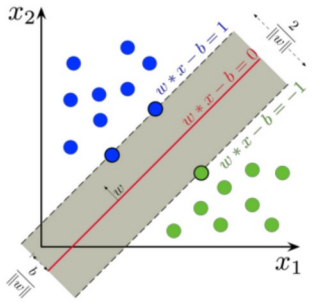
02 How to Learn

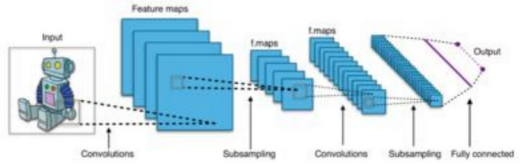
03 Course Logistics

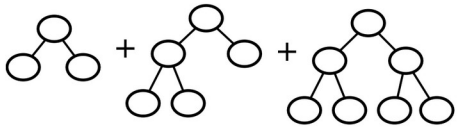
01

Why AI Systems

1958 – 2000: Research







Perceptron Algorithm — 1958

Backprop — 1986

Support Vector Machine (SVM) — 1992

ConvNet — 1998

Gradient Boosting Machine (GBM) — 1999

Slides credits to Tianqi Chen

2000 – 2010: Arrival of Big Data

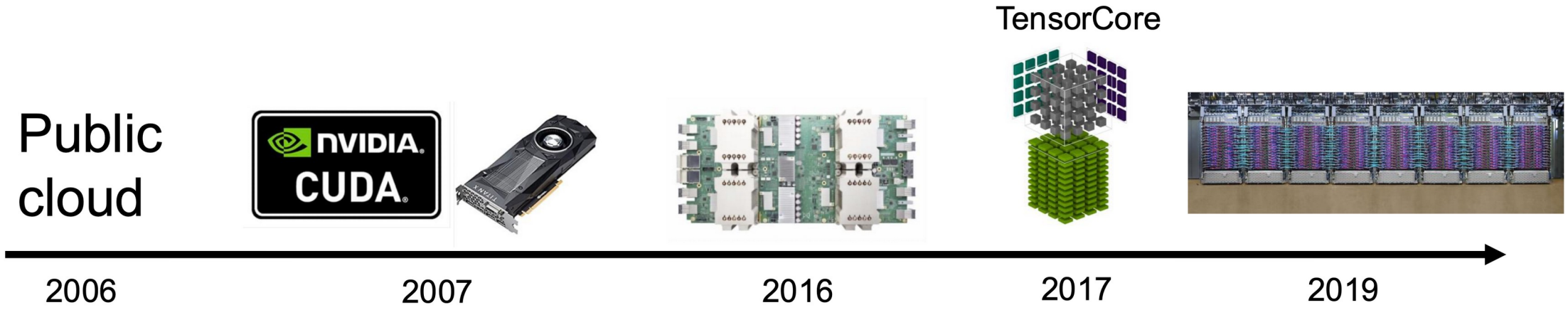
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Data serves as fuel for machine learning models

Slides credits to Tianqi Chen

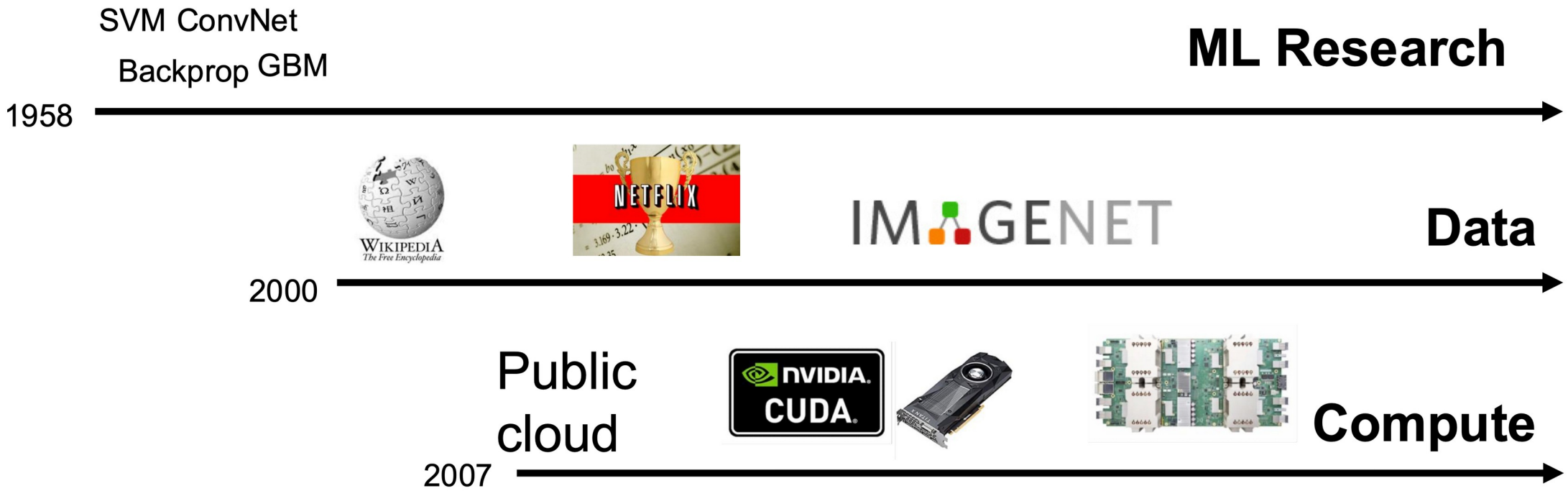
2006–2019: Early Compute Scaling



Compute Scaling

Slides credits to Tianqi Chen

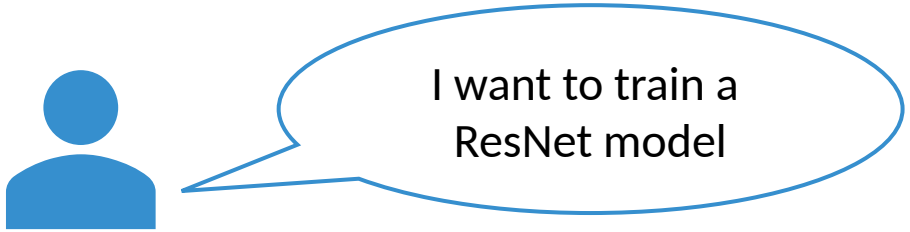
Three Pillars of ML Applications



Slides credits to Tianqi Chen

Research w/o ML System

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Illustrative: ~10k–100k lines across Python/C++/CUDA

Data loader

Model forward pass

Model backward pass

Write optimizer

Forward & backward operators

C++ implementation

CUDA implementation

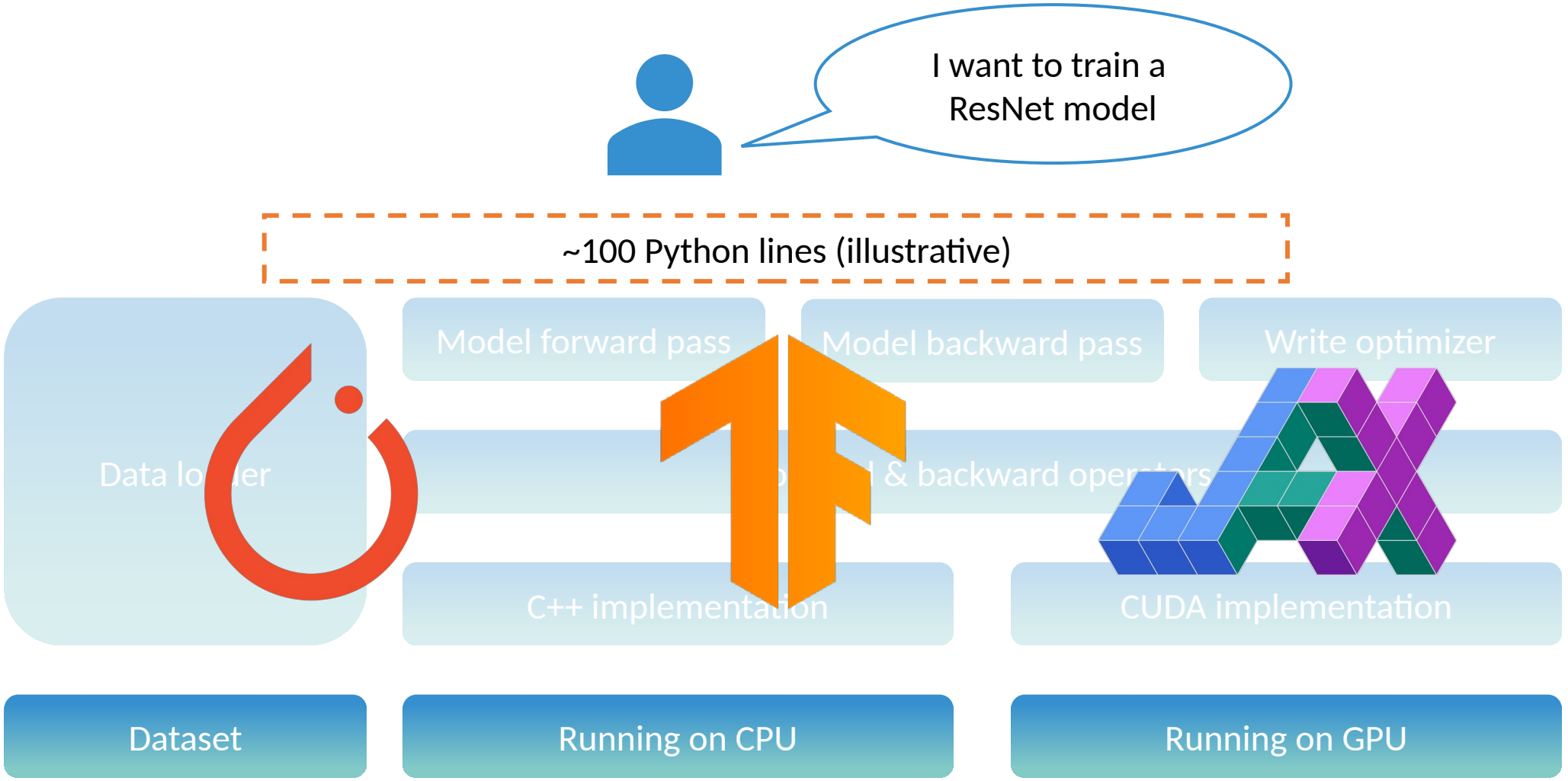
Dataset

Running on CPU

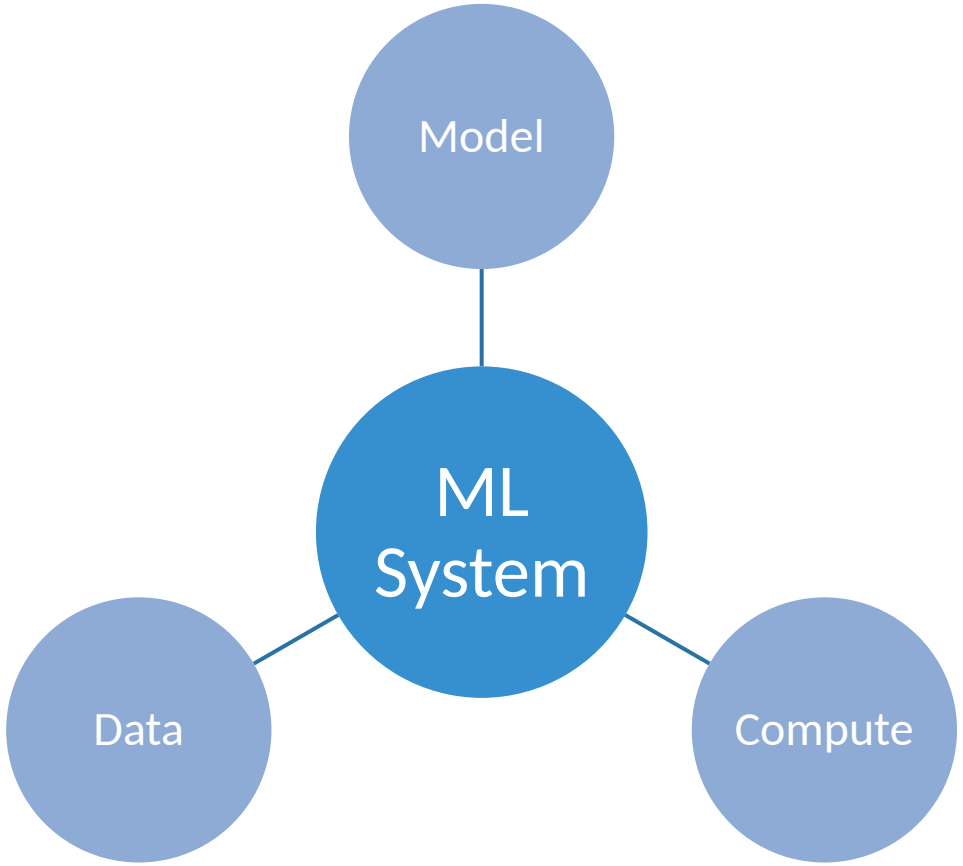
Running on GPU

Research w/ ML System

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ML Systems as a Bridge



ML systems connect models, data, and compute hardware

ML systems support the full machine learning workflow

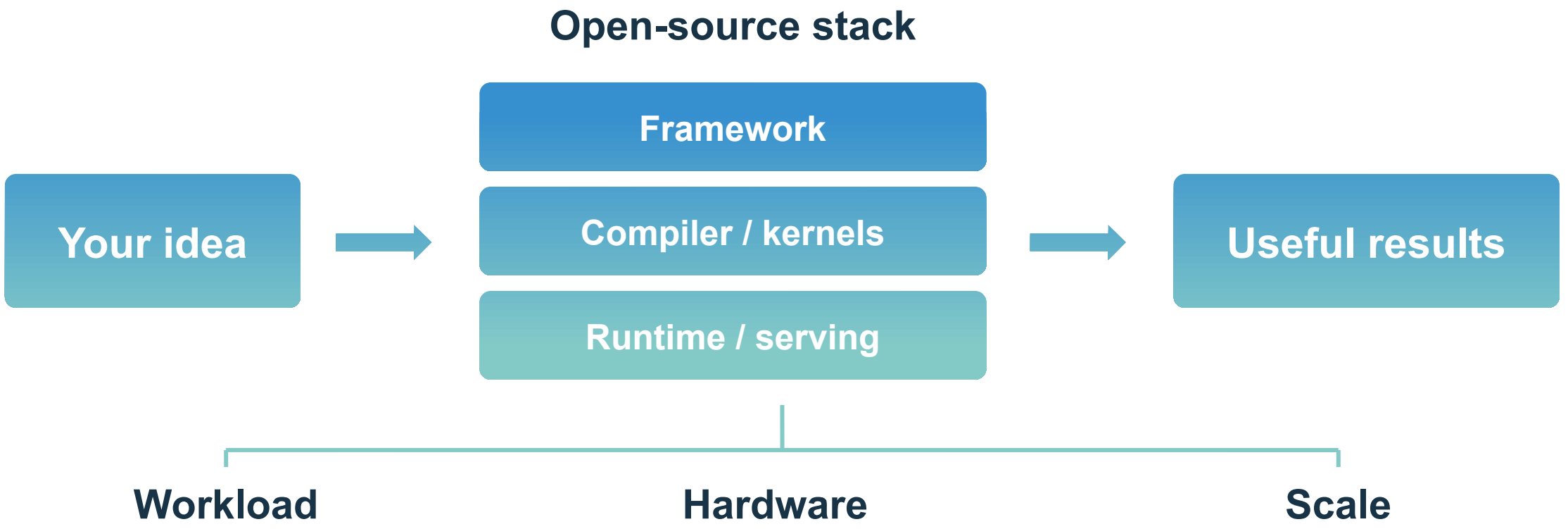
Why Study AI Systems?

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1. AI applications depend on efficient and reliable systems.
2. Understanding the fundamental principles of how AI works facilitates better development of models and algorithms.
3. Efficient software is essential to using domestic AI hardware effectively.
4. Career paths include AI infrastructure, compilers, kernels, and model serving.

Open Source Is a Starting Point

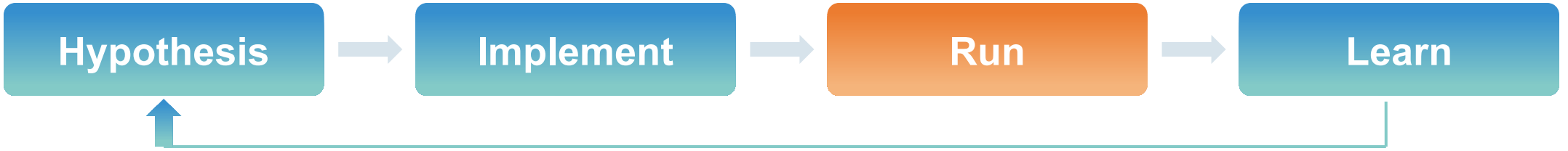
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Choose, configure, diagnose, adapt

Efficiency: Faster Experiment Cycles

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30-day experiment window

3 days / round



10 rounds

1 day / round



30 rounds

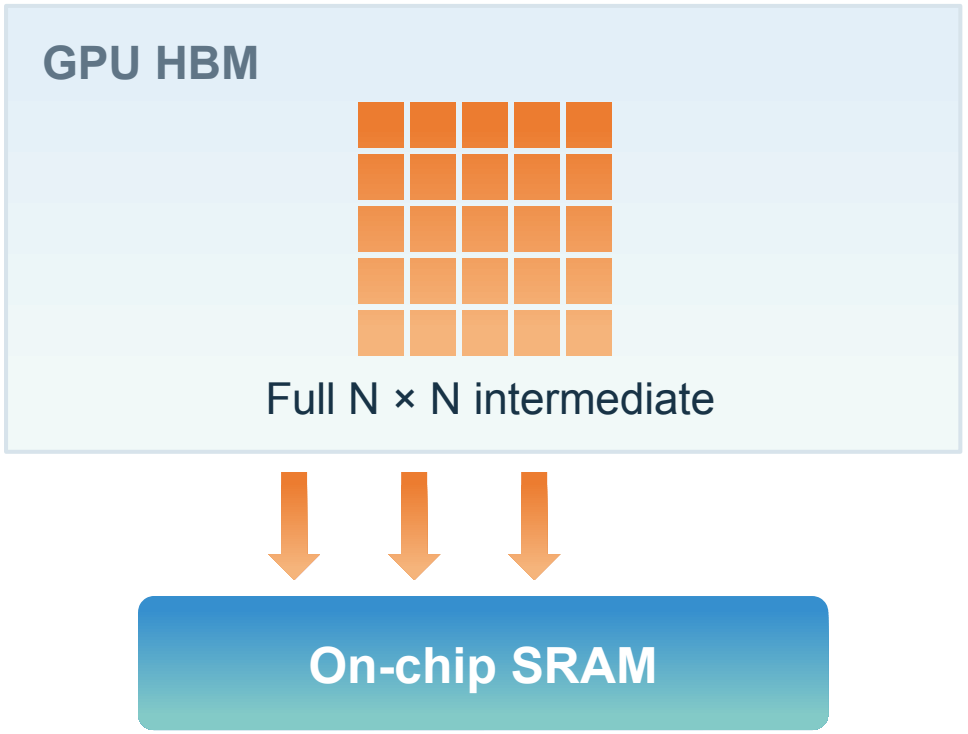
More opportunities to test and refine ideas

Illustrative sequential runs; implementation, queueing, and analysis time excluded.

Efficiency: Algorithms Meet Hardware

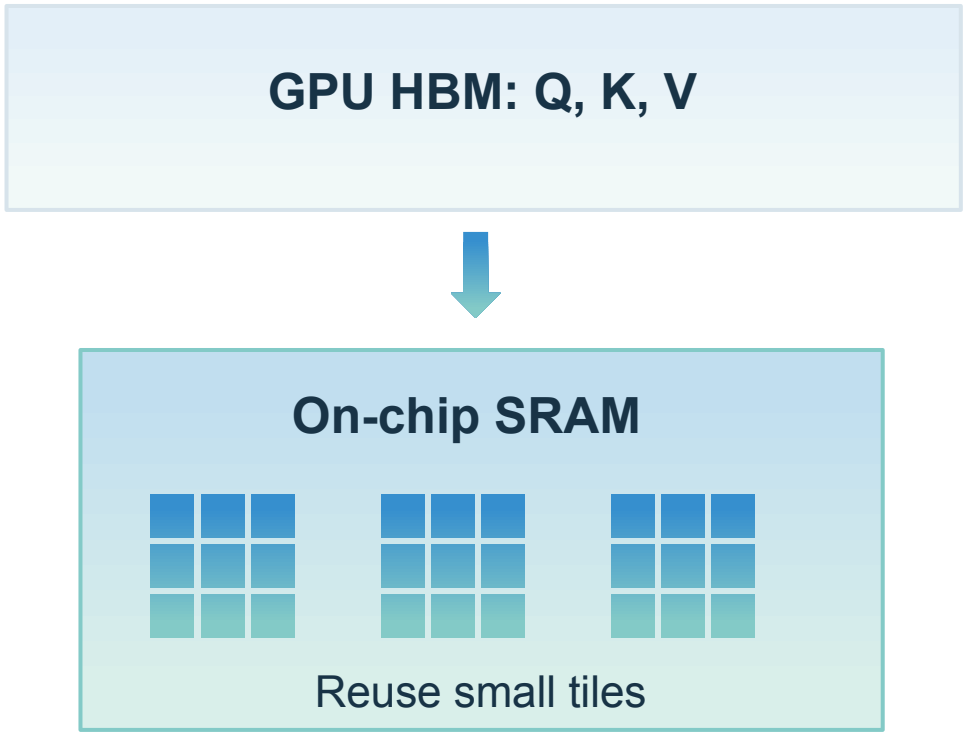
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Conventional attention



Same attention

FlashAttention



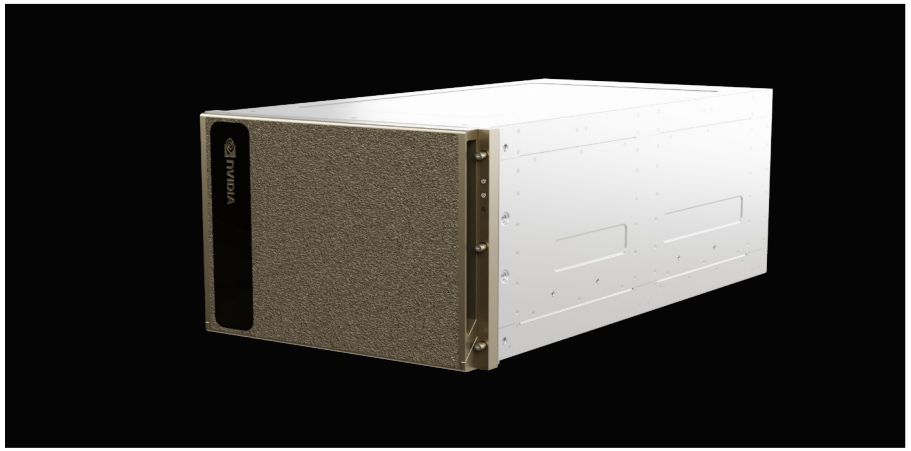
Less data movement

Conceptual memory-flow diagram. FlashAttention, Dao et al., 2022.

Cost: Every Machine-Hour Counts

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H200 server

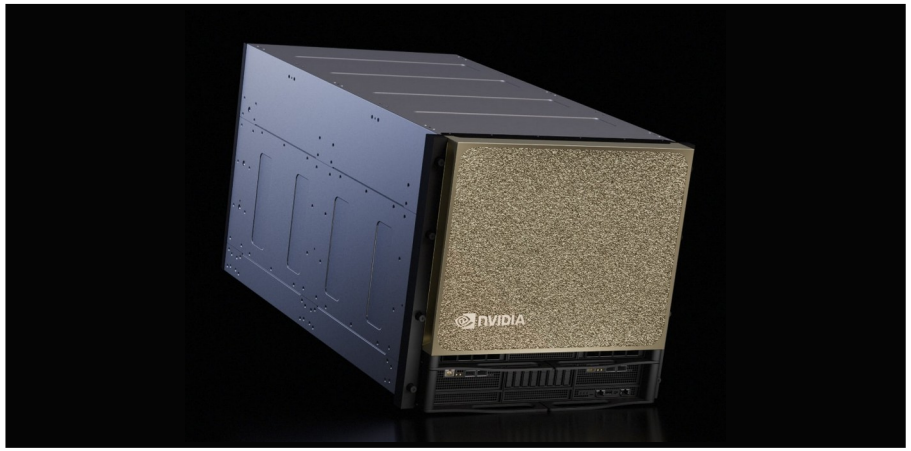


¥5 million

purchase / server

¥120,000 / month

B300 server



¥14 million

purchase / server

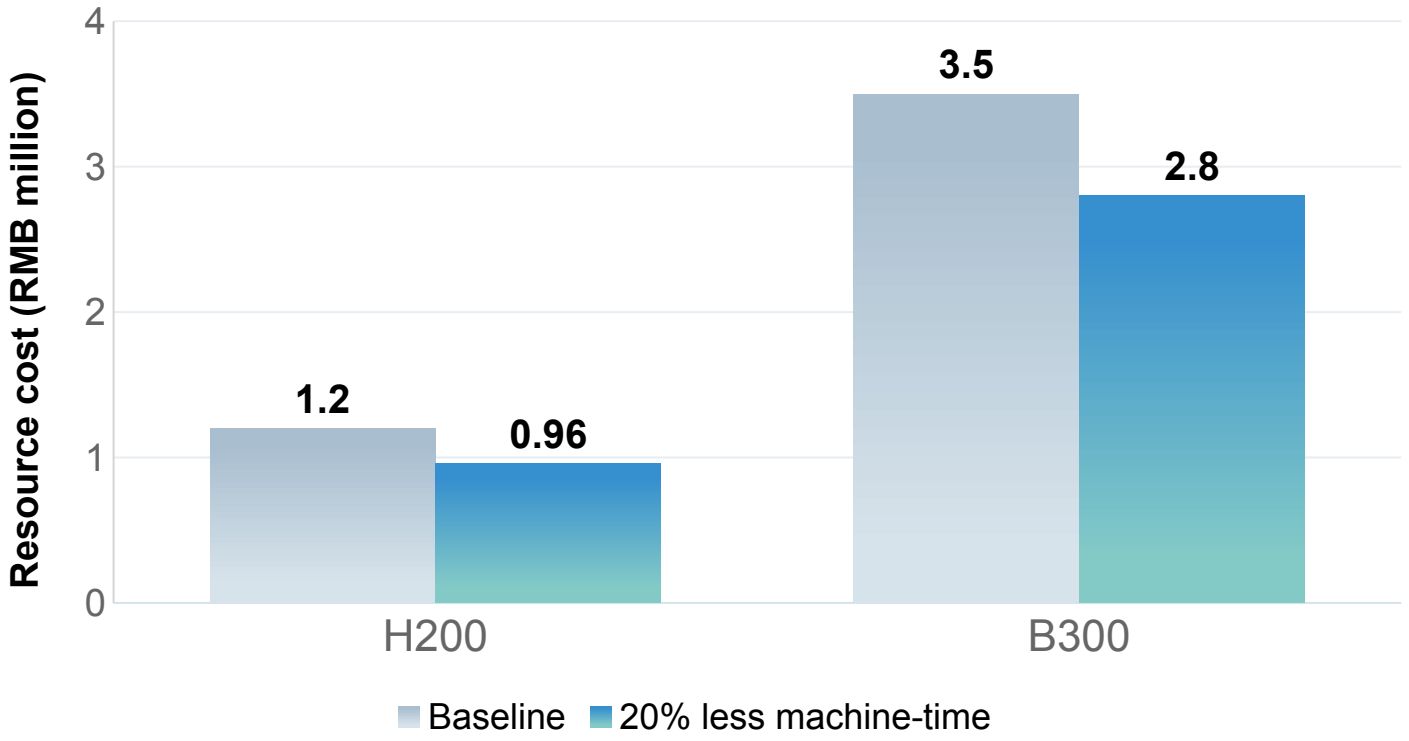
~¥350,000 / month

RMB / server. Instructor-provided price examples; NVIDIA platform images are illustrative.

Cost: The Value of a 20% Reduction

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Same workload, 10 servers, 1 month baseline



Resource value released

¥240k

H200

~¥700k

B300

Resource cost = machines × time × unit price

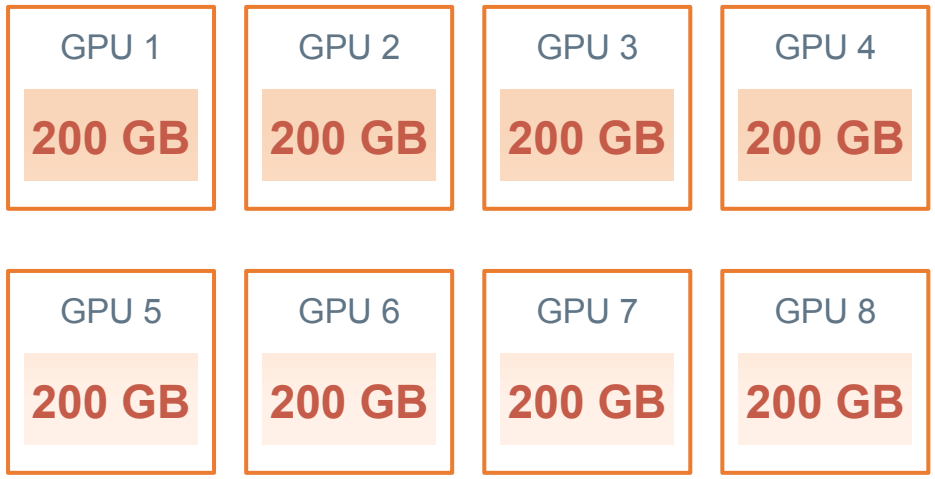
Illustrative resource value. Fixed monthly rental frees capacity; it does not automatically reduce the bill.

Scale: Making Models Fit

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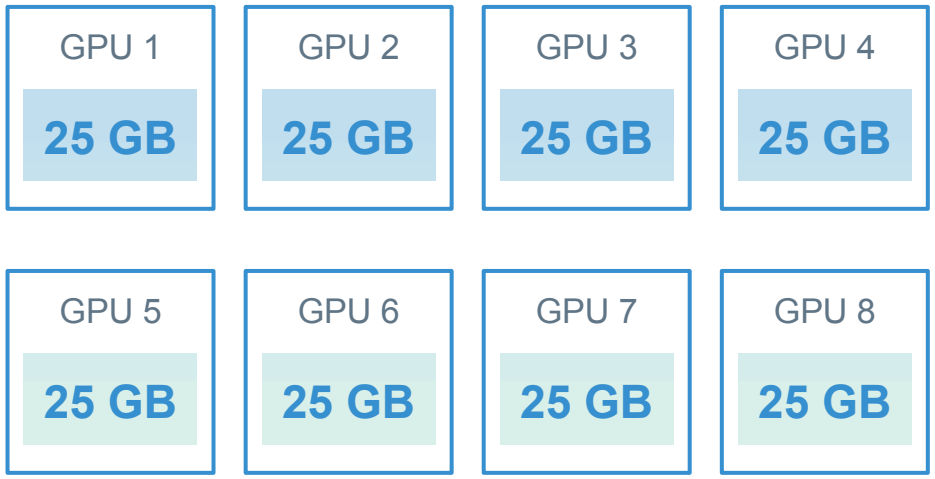
200 GB of training state

Replicate on every GPU



200 GB > 80 GB / GPU

Shard across GPUs



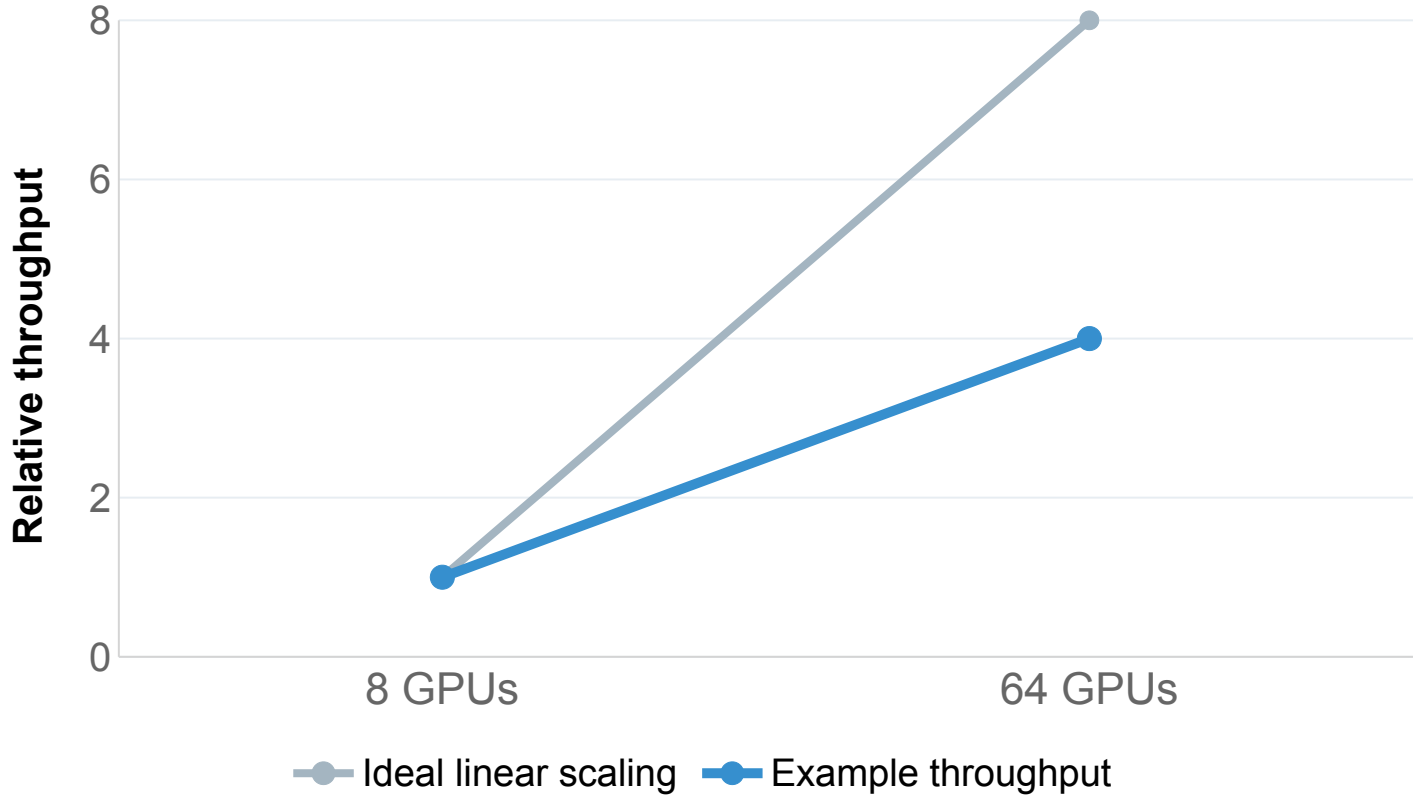
25 GB < 80 GB / GPU

8 × 80 GB = 640 GB aggregate memory. Simplified state-only example; activations and buffers excluded.

Scale: More GPUs, How Much Faster?

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Hypothetical scaling experiment



8× GPUs

4× throughput

50%

scaling efficiency

Possible limits: communication, idle time, load imbalance. Values are illustrative, not measurements.

Why Learn AI Systems?

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01 → **Test ideas faster**

Frameworks · Kernels · Compilation

02 → **Spend compute wisely**

Memory · Training · Serving

03 → **Make scale possible**

Parallelism · Communication

02

How to Learn

Selected Open-Source Milestones

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Project	Scenario	Milestone Date
TensorFlow	DL framework	Nov 2015
PyTorch	DL framework	Jan 2017
Transformers (Hugging Face)	Library	Nov 2018
Megatron-LM	LLM training	Sep 2019
DeepSpeed	LLM training	Feb 2020
FlashAttention	Kernel	May 2022
vLLM	LLM serving	Jun 2023
SGLang	LLM serving	Dec 2023
Mooncake	KVCache management	Jun 2024
verl	RL post-training framework	Oct 2024
DeepEP	Communication library	Feb 2025

Projects evolve rapidly; focus on design principles and performance trade-offs.

No Required Textbook / Recommended Readings

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- Machine Learning / Deep Learning
  - Dive into Deep Learning: https://d2l.ai
- Machine Learning Systems
  - Machine Learning Systems (Foundations / At Scale): https://mlsysbook.ai
  - Microsoft AI-System Education Resource (Chinese): https://github.com/microsoft/AI-System

English materials are recommended

Learning with AI Tools

Use AI to explain concepts, inspect code, and form hypotheses.

Check official documentation for versions and new techniques.

Run a minimal example and verify correctness and performance.

Record the hardware, software version, and input conditions.

03

Course Logistics

Course Instructor

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Siyuan FENG (冯思远)

Assistant professor at Shanghai Innovation Institute

- Ph.D. in computer science, SJTU
- B.Sc. in computer science, ACM class, SJTU
- Research in compilers, AI systems, and AI accelerators
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- Email: syfeng@sii.edu.cn

Course TAs

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殷子珺



于康

Course Schedule

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Fall 2026 | Thursdays 08:55–11:30 | Room 201

16 weeks: Sep 17–Dec 31 | 3 credits / 48 teaching hours

Date (2026)	Teaching plan
Sep 17	Introduction to systems for AI
Sep 24	Automatic differentiation
Oct 01	**[HOLIDAY] National Day — no regular class**
Oct 08	Hardware acceleration
Oct 15	GPU architecture and CUDA programming
Oct 22	NPU architecture and Ascend C programming
Oct 29	Machine learning compilation
Nov 05	Introduction to LLMs and optimizations
Nov 12	Introduction to distributed computing
Nov 19	LLMs: parallelization and training techniques I
Nov 26	LLMs: parallelization and training techniques II
Dec 03	LLMs: parallelization and training techniques III
Dec 10	LLMs: serving techniques I
Dec 17	LLMs: serving techniques II
Dec 24	LLMs: serving techniques III
Dec 31	LLMs: post-training techniques

Holidays: Mid-Autumn Festival, Sep 25–27; National Day, Oct 1–7.
Schedule changes and make-up classes will be announced separately.

Prerequisites

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- Basic linear algebra and calculus
- Python programming and basic tensor operations
- Ability to read, debug, and verify code, including AI-generated code
- (Optional) C++ / CUDA programming
- (Optional) Computer architecture and networks

Grading

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[10%] Attendance

1st unexcused absence: warning
2nd unexcused absence: attendance score halved
3rd unexcused absence: attendance score becomes 0

[40%] Assignments

Two assignments, 20% each
Topics and deadlines will be announced later

[50%] Final Exam

Closed-book; no AI tools or electronic devices
Exam date and arrangements will be announced later

Attendance and Assignment Waivers

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No full course exemption is offered this year.

Students who have previously taken this course may apply for waivers of attendance and regular assignments.

All students must take the final exam.

Please contact Siyuan FENG to apply: syfeng@sii.edu.cn

Course Updates and Feedback

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- This course is updated each year. Feedback on course materials is welcome.
- AI systems and open-source projects continue to evolve rapidly.
- Changes to course materials, scheduling, and make-up classes will be announced separately.

Policy for the Use of AI Tools

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Final exam: closed-book; no AI tools or electronic devices.

AI tools are permitted in all other course activities.

The use of any form of Artificial Intelligence (AI) tools is permitted and encouraged in this course to support learning and research. Students are not required to declare or cite the use of AI tools in their submissions, including but not limited to assignments, projects, and reports.

Students are held fully accountable for all submitted materials, including but not limited to code, experimental data, and technical reports. Should the use of AI tools result in any adverse consequences—such as the submission of malicious or destructive code, data fabrication, or other forms of academic misconduct including plagiarism or excessive similarity to other works—the student who made the submission will bear sole responsibility.

Acknowledgement

The development of this course, including its structure, content, and accompanying presentation slides, has been significantly influenced and inspired by the excellent work of instructors and institutions who have shared their materials openly. We wish to extend our sincere acknowledgement and gratitude to the following courses, which served as invaluable references and a source of pedagogical inspiration:

- Machine Learning Systems[15-442/15-642], by **Tianqi Chen** and **Zhihao Jia** at **CMU**.
- Advanced Topics in Machine Learning (Systems)[CS6216], by **Yao Lu** at **NUS**

While these materials provided a foundational blueprint and a wealth of insightful examples, all content herein has been adapted, modified, and curated to meet the specific learning objectives of our curriculum. Any errors, omissions, or shortcomings found in these course materials are entirely our own responsibility. We are profoundly grateful for the contributions of the educators listed above, whose dedication to teaching and knowledge-sharing has made the creation of this course possible.

System for Artificial Intelligence

Thanks

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